

Name: *Ali Syed*

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Work Auth: US Citizen



Professional Summary:

- Seasoned Data Scientist with extensive 9+ experience in designing and deploying machine learning models for complex healthcare and financial use cases
- Strong background in statistical analysis, predictive modeling, and experimental design to drive data-informed decision-making
- Proficient in Python, R, and SQL, with a focus on data wrangling, feature engineering, and advanced data visualization
- Demonstrated expertise in AI/ML frameworks (e.g., TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM) to solve classification, regression, and time-series forecasting problems
- Skilled in building end-to-end data pipelines, including data ingestion, cleaning, transformation, and model deployment in production environments
- Hands-on experience with MLOps best practices, employing CI/CD pipelines, containerization (Docker), and orchestration (Kubernetes)
- Adept at cloud computing (AWS/Azure) for large-scale data processing, model training, and scalable architecture design
- Thorough understanding of healthcare data privacy and security guidelines, including HIPAA compliance and IRB protocols
- Proven ability to lead cross-functional teams, collaborate with stakeholders, and present insights to both technical and non-technical audiences
- Experience in deploying Natural Language Processing (NLP) and computer vision solutions for diverse healthcare applications
- Proficient in time-series forecasting, anomaly detection, and real-time analytics for proactive decision-making
- Adept at agile methodologies and sprint planning to ensure timely delivery of high-impact data science projects
- Strong background in data governance and establishing processes for ensuring data quality, lineage, and metadata management
- Emphasis on continuous learning and staying updated with emerging AI technologies, including generative modeling and reinforcement learning
- Recognized for strong analytical thinking, problem-solving skills, and ability to foster a culture of innovation within data science teams
- Expertise knowledge on Object Oriented Programming (OOPS) Concepts including Inheritance, Abstraction, Encapsulation, and Polymorphism.
- Launching Amazon EC2 Cloud Instances using Amazon Web Services (Linux/ Ubuntu) and configuring launched instances concerning specific applications.
- Rapid provisioning of pre-configured Linux instances on demand in DEV/IT, UAT, and Production environment through AWS (EC2).

Technical Skills

Operating Systems:	Windows,Mac, UNIX, Linux.
Programming Languages:	C, Python, R, SQL, Bash, Typescript, JavaScript, Go Lang, Pandas

Machine Learning	Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch
Data Analysis	Pandas, NumPy, SciPy, statsmodels, matplotlib, Seaborn, Plotly, Tableau, Power BI, Spark, Hadoop, Hive, Kafka, Airflow, Kubeflow, MLflow, Jenkins, GitHub Actions, Docker, Kubernetes
Cloud Tools	AWS (EC2, S3, Lambda, SageMaker), Azure (Databricks, Azure ML)
Unit Testing Frameworks:	MS-Test, Nunit, Jasmine, Karma
Dependency Injection Tools:	Structure Map, Ninject, Unity Framework
Database Products:	MS-SQL Server, Oracle, DB2, MySQL, T-SQL, Sybase, Oracle 11G, SQL Server 2005/2008/2012, MS Access
SOA:	Web Services, SOAP, WSDL, WCF, XSD, XPath, REST, OData
Version Control:	SVN, TFS, GIT, Accu Rev, MKS
ERP:	SAP HANA, SAP HANA Cloud Integration, SAP BODS
Object-Oriented Methodologies:	UML, Design Patterns
Source Repositories:	Git Hub, TFS, VSS
Modeling and Design Tools:	Microsoft Visio, IBM Clear case, Rational Rose, UML

Professional Experience:

Noridian Healthcare, HighPoint, NC

July 2018 – Till Date

Role: Senior Data Scientist

Responsibilities:

- Led the development of predictive models to identify high-risk patient populations, reducing hospital readmission rates by over 15%
- Deployed AI-driven anomaly detection systems for early diagnosis of critical conditions, improving patient outcomes and resource allocation
- Implemented privacy-preserving machine learning techniques to ensure compliance with healthcare data regulations (HIPAA, IRB)
- Designed and optimized ETL pipelines to handle large volumes of patient data from multiple sources, improving data integrity by 25%
- Utilized time-series analysis to forecast patient admissions and staffing needs, cutting operational costs by 10%
- Spearheaded an initiative to integrate electronic health records (EHR) data with wearable device insights for comprehensive patient monitoring
- Created interactive dashboards using Tableau and Plotly to communicate actionable insights to clinicians and administrative stakeholders
- Automated feature engineering workflows, cutting data preparation time by 40% and enabling faster model iterations
- Mentored a team of junior data scientists on advanced machine learning techniques and code optimization best practices
- Orchestrated A/B testing frameworks for clinical trials, enabling robust evaluation of new treatments and interventions

- Collaborated with cross-functional teams (Medical, IT, Legal) to define project scopes, data security protocols, and success metrics
- Built custom Python libraries for healthcare data preprocessing, expediting the analytical process and promoting code reusability
- Introduced cloud-based model deployment (AWS) with automated scaling, ensuring consistent performance during peak usage
- Led root cause analyses to mitigate data quality issues and implement data governance strategies across departments
- Pioneered the use of deep learning for medical image classification, improving diagnostic accuracy for complex conditions
- Experience in deploying Natural Language Processing (NLP) and computer vision solutions for diverse healthcare applications
- Proficient in time-series forecasting, anomaly detection, and real-time analytics for proactive decision-making
- Adept at agile methodologies and sprint planning to ensure timely delivery of high-impact data science projects
- Strong background in data governance and establishing processes for ensuring data quality, lineage, and metadata management
- Emphasis on continuous learning and staying updated with emerging AI technologies, including generative modeling and reinforcement learning
- Developed various stored procedures and views for data retrieval from database, to generate different reports using SQL Reporting Services (SSRS).

Environment: Airflow, Apache Spark, AWS (EC2, S3, Lambda, Redshift), Azure ML, Docker, FastAPI, Flask, Git, GitHub, Jenkins, Keras, Kubernetes, LightGBM, MLflow, Power BI, PostgreSQL, Python 3.x, PyTorch, R, scikit-learn, SQL Server 2016+, Tableau, TensorFlow, XGBoost

Paychex, Greensboro, NC

Feb 2016 – Apr 2019

Role: Sr. Data Scientist

Responsibilities:

- Developed machine learning solutions to optimize payroll processes, reducing errors and processing time by 20%
- Built predictive analytics models for revenue forecasting and resource planning across multiple business units
- Deployed a fraud detection system leveraging advanced ensemble methods, cutting fraudulent transactions by 30%
- Established end-to-end MLOps processes, integrating Git, Docker, Jenkins, and Kubernetes for seamless model deployment
- Implemented natural language processing (NLP) techniques to automate customer support ticket classification, improving response times
- Conducted customer segmentation analysis using clustering algorithms, enabling targeted marketing strategies and boosting engagement
- Oversaw real-time analytics initiatives, leveraging Spark and Kafka to process high-velocity payment data
- Designed a feature store for consistent feature definitions and streamlined offline/online model serving
- Played a pivotal role in data architecture improvements by migrating legacy systems to more scalable, cloud-based solutions
- Streamlined ETL processes with Apache Airflow, resulting in a 40% decrease in data latency for business dashboards
- Advised business leaders on ROI-driven project prioritization, balancing immediate needs with long-term data strategy

- Championed an Agile environment, coaching scrum teams to deliver iterative improvements and maintain stakeholder alignment
- Experimented with reinforcement learning approaches to optimize scheduling and allocation of limited resources
- Coached and trained data science interns in core data mining and statistical modeling practices
- Liaised with legal, compliance, and security teams to ensure data protection standards (GDPR, SOC 2) were consistently met
- Strong background in statistical analysis, predictive modeling, and experimental design to drive data-informed decision-making
- Proficient in Python, R, and SQL, with a focus on data wrangling, feature engineering, and advanced data visualization
- Demonstrated expertise in AI/ML frameworks (e.g., TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM) to solve classification, regression, and time-series forecasting problems
- Skilled in building end-to-end data pipelines, including data ingestion, cleaning, transformation, and model deployment in production environments
- Hands-on experience with MLOps best practices, employing CI/CD pipelines, containerization (Docker), and orchestration (Kubernetes)
- Adept at cloud computing (AWS/Azure) for large-scale data processing, model training, and scalable architecture design
- Supporting Various Web applications, SSRS, SSIS, SQL Server, and MS Access, across sites
- Providing the Estimation for new developments and enhancements.
- Involved in Migration/Upgrade of Data Base servers

Environment: Python 3.x, R, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, Keras, Docker, Kubernetes, Jenkins, Git, GitHub, Azure ML, AWS (EC2, S3, Lambda, Redshift), SQL Server 2016+, PostgreSQL, Tableau, Power BI, Apache Spark, Airflow, MLflow, .NET Core, FastAPI, Flask

Education:

- Bachelors in Computer Science, ECPI University, North Carolina, 2016